Craftax runs compared: achievements and what unlocks them

The goal-manager family (v1-v1.3, v2) and the plain baselines: 1.1M env steps each, size50m, scored on the same 30 fixed worlds with tools/death_eval.py. Honest runs first; [masked] runs read the game's action rules and count as cheating. Rows with several training seeds show seed 0. Updated 2026-11-08.

Every number is the mean episode-final achievement count over 30 evaluation episodes, ± its standard error over those worlds; unlock rates are the share of episodes in which an achievement fired; return is the mean raw Craftax return as a % of the maximum (226). Every checkpoint is scored on the same fixed worlds (env.reseed, audit B05), so comparisons are paired world by world. The error bars cover the worlds only, not training seeds; a 30-world score moves by about ±0.55 between world sets, so a gap under ~1.1 between two single runs is not evidence.

Honest runs (no rule mask): plain 4.77 (two seeds: 4.77 / 4.77); plain + RSSM-2 memory + the v1.2 manager 5.77 (seed 1: 5.30; two-seed mean 5.53, +0.77 ± 0.22 over plain, +0.20 ± 0.24 over RSSM-2 alone, which scores 5.33 on family/rssm2); fixed v1.2 without the mask 6.03 (6.13 ± 0.22 over 60 worlds, +0.33 ± 0.40 over the honest map it is built on). v2 without the mask (v2_norules) was trained but not scored: it ends training at 4.78 achievements per episode (last 100k steps), below plain (5.43).

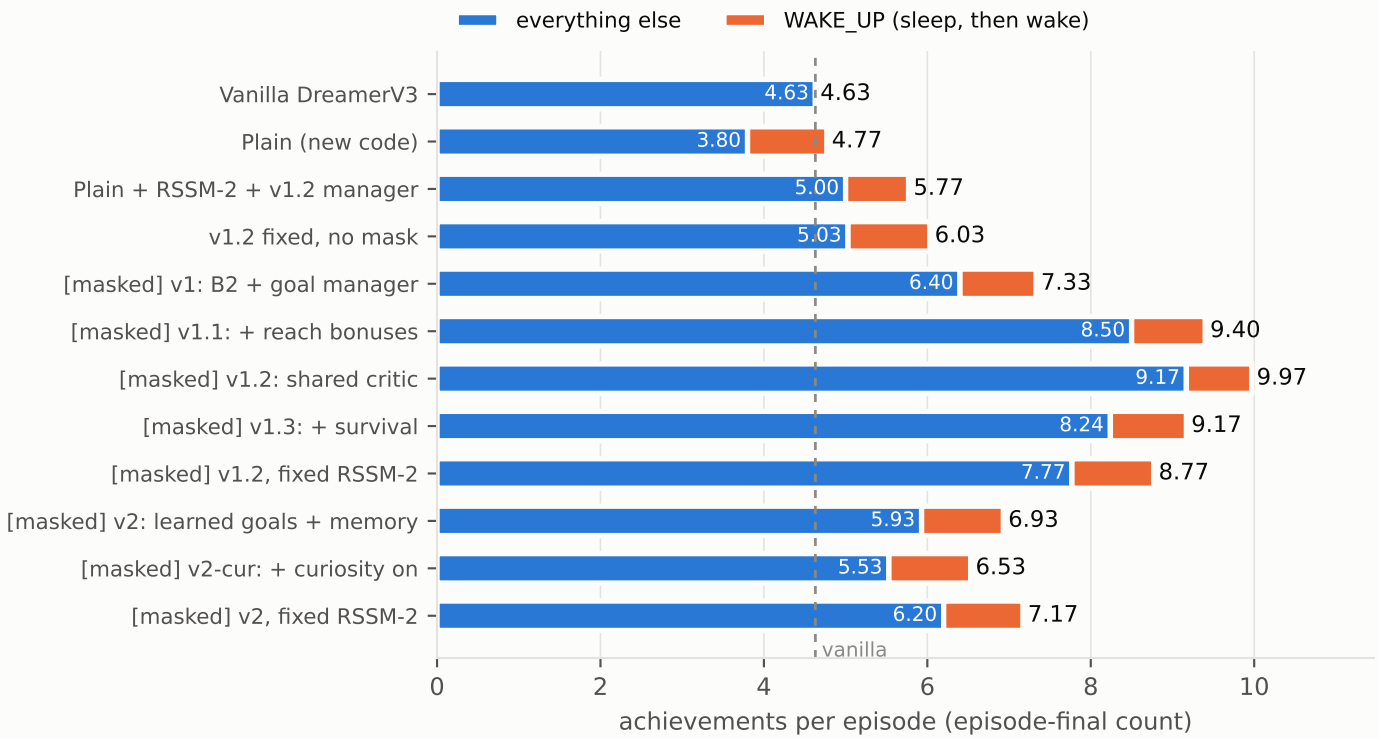
Masked runs (counted as cheating; B2 rule mask on): v1 7.33, v1.1 9.40, v1.2 9.97, v1.3 9.17 (old code); v1.2 retrained on the fixed RSSM-2 code 8.77 (three seeds 8.77 / 8.83 / 8.73; 9.03 over 60 worlds); v2 6.93, v2-cur 6.53, v2 fixed 7.17 (seeds 7.17 / 7.83 / 6.07). The mask is worth +2.90 ± 0.26 under the v1.2 manager (60 worlds), as much as without a manager.

run	ach. ± SE	w/o wake	life	return	table	pickaxe	sword	stone	drink
Vanilla DreamerV3	4.63 ± 0.22	4.63	295	1.65%	63%	23%	3%	0%	77%
Plain (new code)	4.77 ± 0.28	3.80	265	1.71%	50%	3%	10%	0%	73%
Plain + RSSM-2 + v1.2 manager	5.77 ± 0.24	5.00	255	2.15%	97%	37%	17%	3%	83%
v1.2 fixed, no mask	6.03 ± 0.33	5.03	275	2.27%	70%	33%	13%	20%	70%
[masked] v1: B2 + goal manager	7.33 ± 0.53	6.40	256	2.85%	90%	57%	27%	37%	67%
[masked] v1.1: + reach bonuses	9.40 ± 0.54	8.50	274	3.76%	93%	80%	43%	80%	90%
[masked] v1.2: shared critic	9.97 ± 0.36	9.17	262	4.04%	100%	93%	50%	87%	67%
[masked] v1.3: + survival	9.17 ± 0.49	8.24	294	3.66%	97%	80%	43%	67%	97%
[masked] v1.2, fixed RSSM-2	8.77 ± 0.52	7.77	319	3.48%	97%	67%	33%	67%	73%
[masked] v2: learned goals + mem	6.93 ± 0.36	5.93	270	2.67%	93%	37%	20%	27%	73%
[masked] v2-cur: + curiosity on	6.53 ± 0.22	5.53	319	2.50%	100%	23%	30%	7%	80%
[masked] v2, fixed RSSM-2	7.17 ± 0.44	6.20	276	2.77%	90%	53%	23%	30%	87%

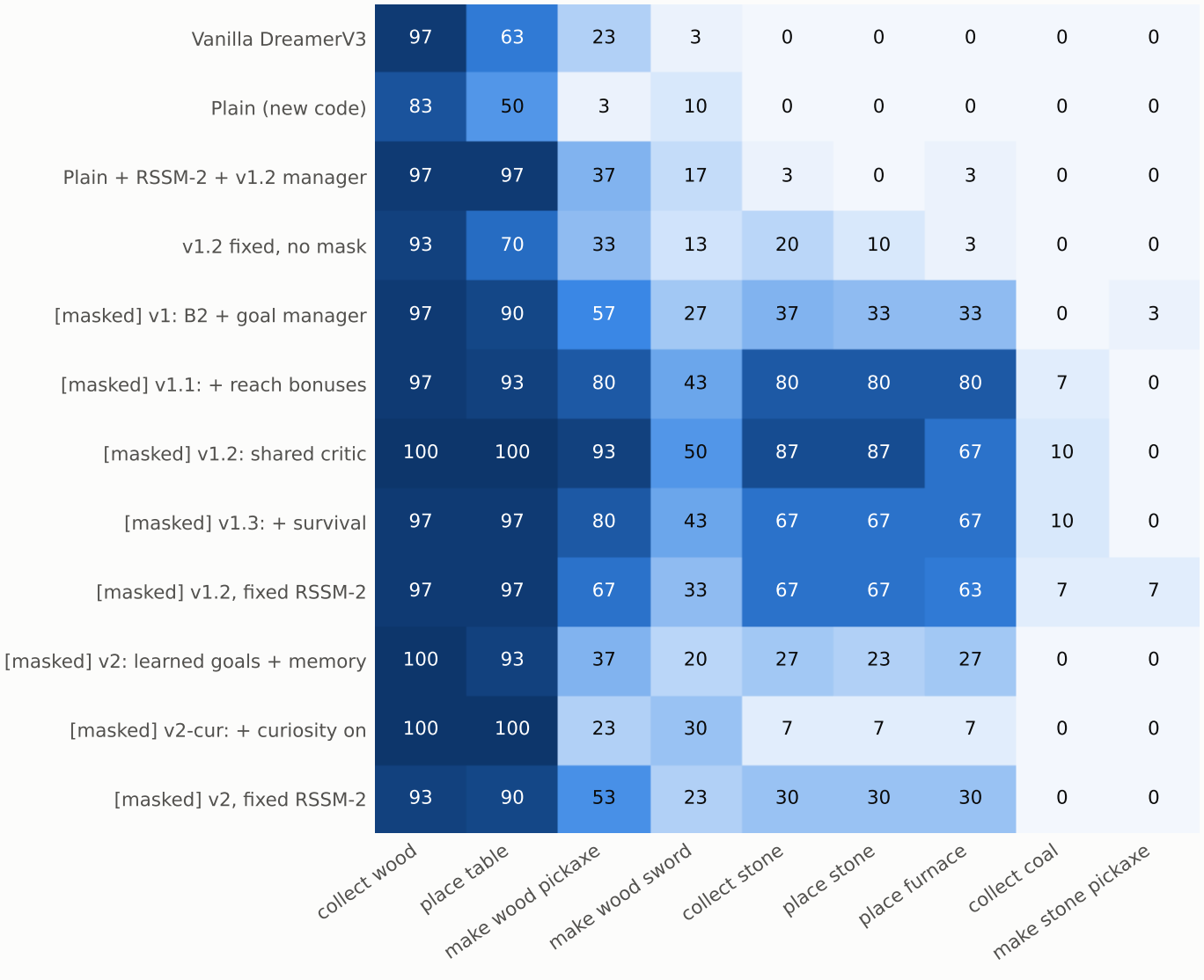
ach. ± SE = mean episode-final achievements ± standard error over the 30 worlds (one training seed). w/o wake = the same minus WAKE_UP. life = mean lifespan in steps. return = raw Craftax return, % of 226. Rates are % of episodes. Vanilla DreamerV3: the original checkpoint (code before the 2026-10 fixes), re-scored. Plain: main's code. +manager: plain + RSSM-2 as a memory + the v1.2 manager (13 named goals, shared critic, hold 32), no mask, no shaping. v1.2 no mask: fixed v1.2 (map-target RSSM-2, potential shaping) minus the rule mask and the validity flags. [masked] rows use the B2 rule mask: v1-v1.3 on the old code (RSSM-2 bugs, docs/design-manager.md); v2 and v2-cur learned goals, curiosity, RSSM-2 memory (docs/design-v2.md); "fixed RSSM-2" rows were retrained on the code of the 2026-10-11 audit.

Achievements per run, and the early tech tree

Mean episode-final achievements per run, split into WAKE_UP and the rest; unlock rate of the early tech-tree rungs



Unlock rate (% of episodes), early tech tree in dependency order



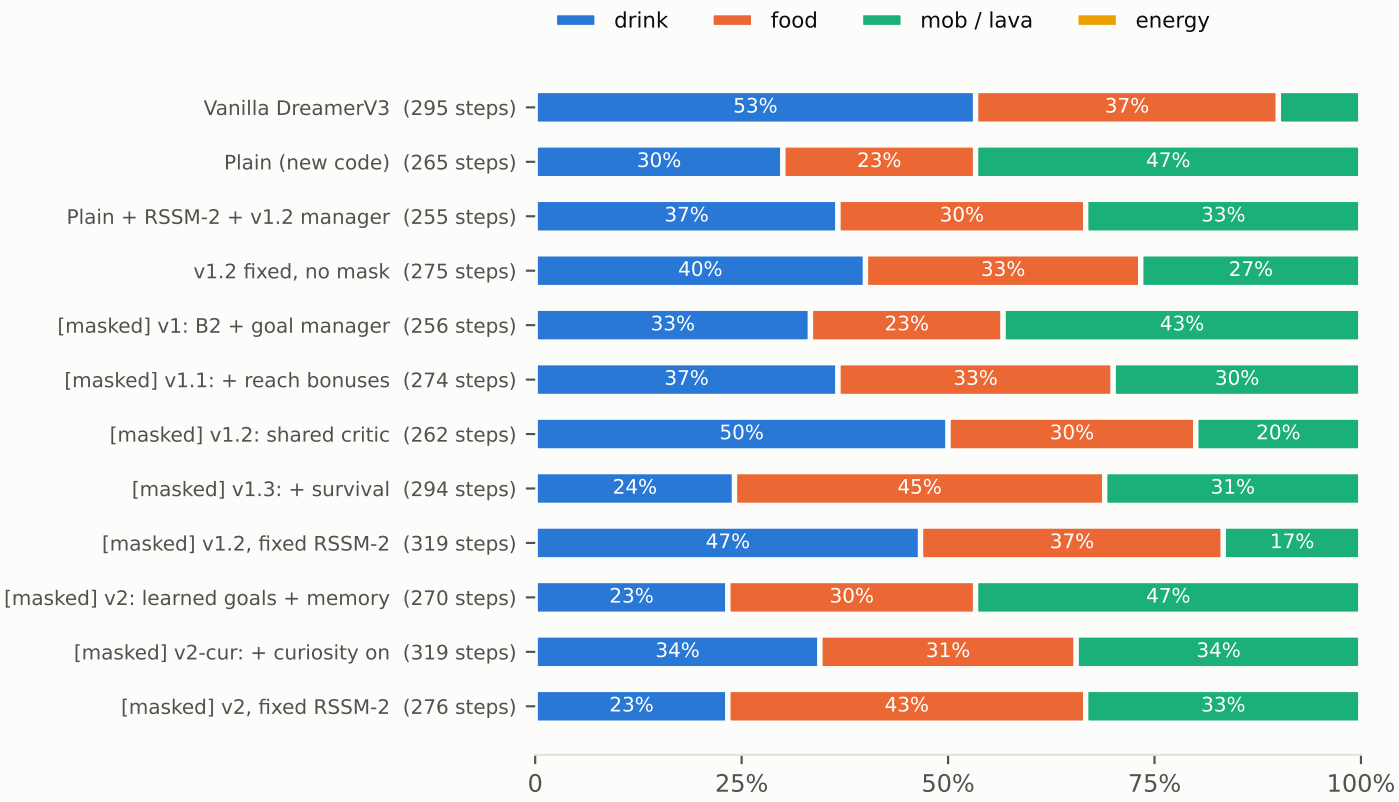
Every achievement any run unlocked

% of evaluation episodes in which each achievement fired. The other 47 of 67 were never unlocked by any run.



Cause of death and lifespan

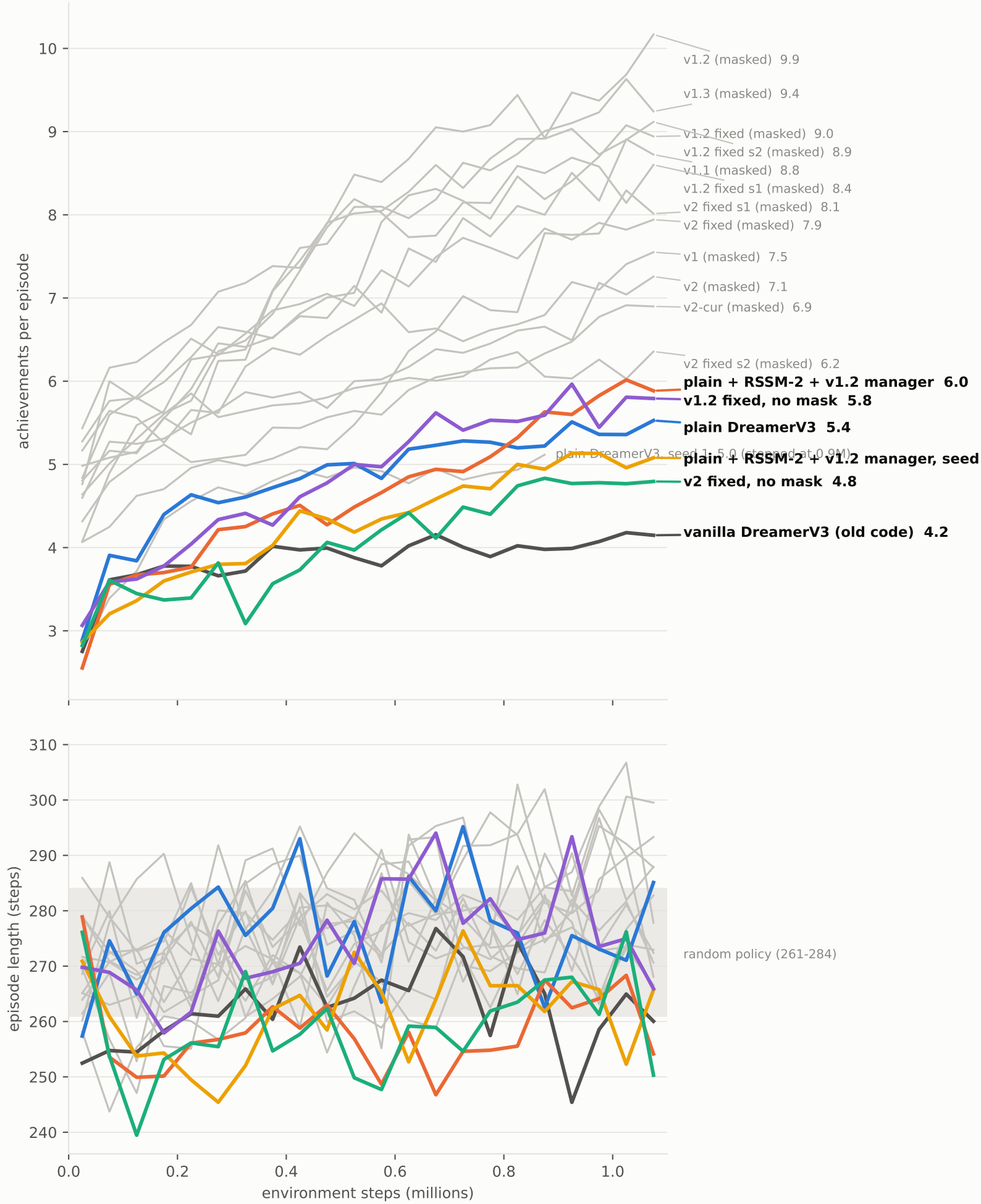
Cause of death (share of evaluation episodes) and lifespan; and, if the results list one, how much a run moves between its own checkpoints



Random policy lives 261-284 steps; doing nothing lives 333.

Training curves: goal manager vs plain

Rebuilt from each run's replay; 50k-step bins; value = mean of the last 100k steps.



What the results say, and what would settle the rest

1. Without the mask, the manager adds little

On plain DreamerV3 the v1.2 manager (with the RSSM-2 memory it reads) is +0.77 ± 0.22 over plain on two seeds, but only +0.20 ± 0.24 over RSSM-2 alone, and the second seed ended training below plain. Fixed v1.2 without the mask is +0.33 ± 0.40 over the honest map underneath it. No honest manager run makes a stone tool.

2. The masked scores are largely the mask's

Every v1.x and v2 headline score has the B2 rule mask, which reads the game's action rules and now counts as cheating. Removing it drops fixed v1.2 from 9.03 to 6.13 over 60 worlds (-2.90 ± 0.26) and v2 by about 2.6 achievements in training (v2_norules). Among masked runs, v1.2 leads B2 by +0.50 ± 0.25 on 60 worlds and is level with it in training.

3. Old v1.2 (9.97) was partly luck

Rescored on a second set of 30 worlds it gives 9.50 (9.73 over 60). Its fixed-code reruns average 9.03. Part of the old lead may come from the pre-fix RSSM-2 leak (training saw up to 7 future steps, and the manager plans from those states); one old seed cannot tell (docs/ablations.md).

4. What would settle the rest

A score for v2_norules, more seeds of v12_nomask and abl_manager, and 60-world scores throughout. The next version of the manager is on family/next (v1.2's named goals on v2).